



Session 4: Takeoff Performance & Optimum Altitude & Landing Performance

Date: 04 September 2020

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Takeoff Performance

Takeoff Performance

- **Regulatory Requirements**
- **Wet, Slippery, and Contaminated Runways**
- **Flap and Speed Selection**
- **Reduced and Derated Thrust**
- **Alternate Forward CG Limits**
- **OPT Takeoff Calculations**

Takeoff Performance

Regulatory Requirements for Takeoff

① Field Length Requirement

- 35 feet with engine failure at V_1
- 35 feet all engine distance plus 15%
- Stop with event recognized at V_1

② Obstacle clearance

- Net flight path must clear obstacle by 35 feet.
- Net = gross path degraded by 0.8% gradient for 2-eng aircraft

③ Climb Gradient Requirement

- 1st Segment - Positive climb gradient
- 2nd Segment - 2.4% climb gradient
- Final Segment - 1.2%

④ Tire speed

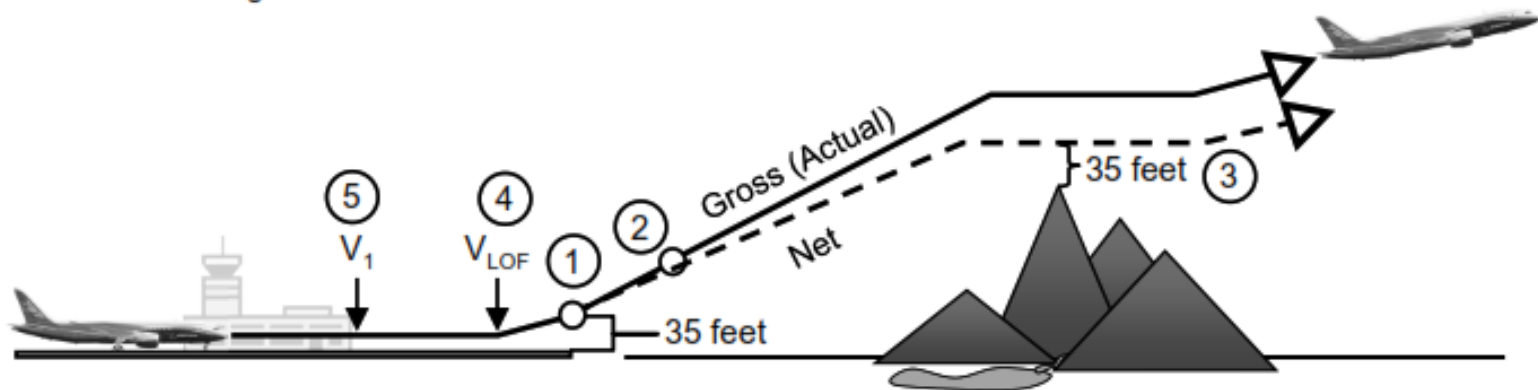
- The ground speed at the lift-off speed (V_{LOF}) must not exceed the tire speed limit (235 mph on 787 models)

⑤ Brake energy

- V_1 must be less than or equal to V_{MBE}

⑥ Return-to-Land

- The airplane must be able to meet the same minimum go-around gradients that are required for dispatch landing performance

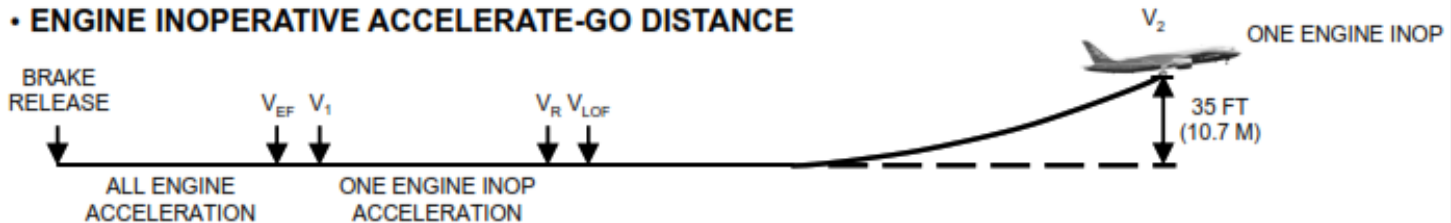


Takeoff Performance

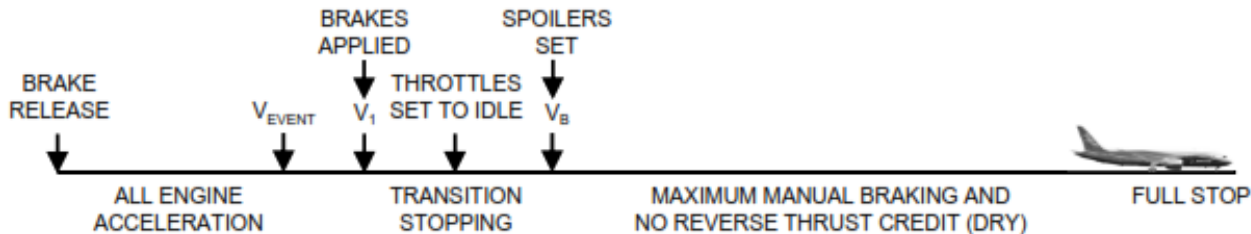
Dry Runway Field Length Requirements

The takeoff field length required is the longest of:

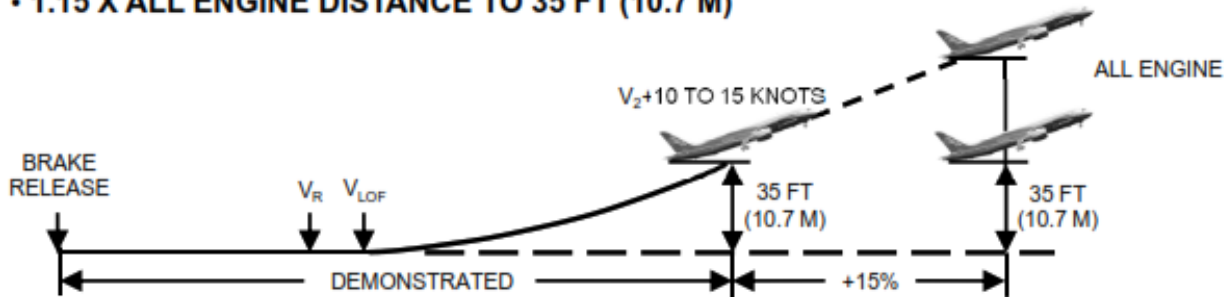
- **ENGINE INOPERATIVE ACCELERATE-GO DISTANCE**



- **ACCELERATE-STOP DISTANCE**



- **1.15 X ALL ENGINE DISTANCE TO 35 FT (10.7 M)**

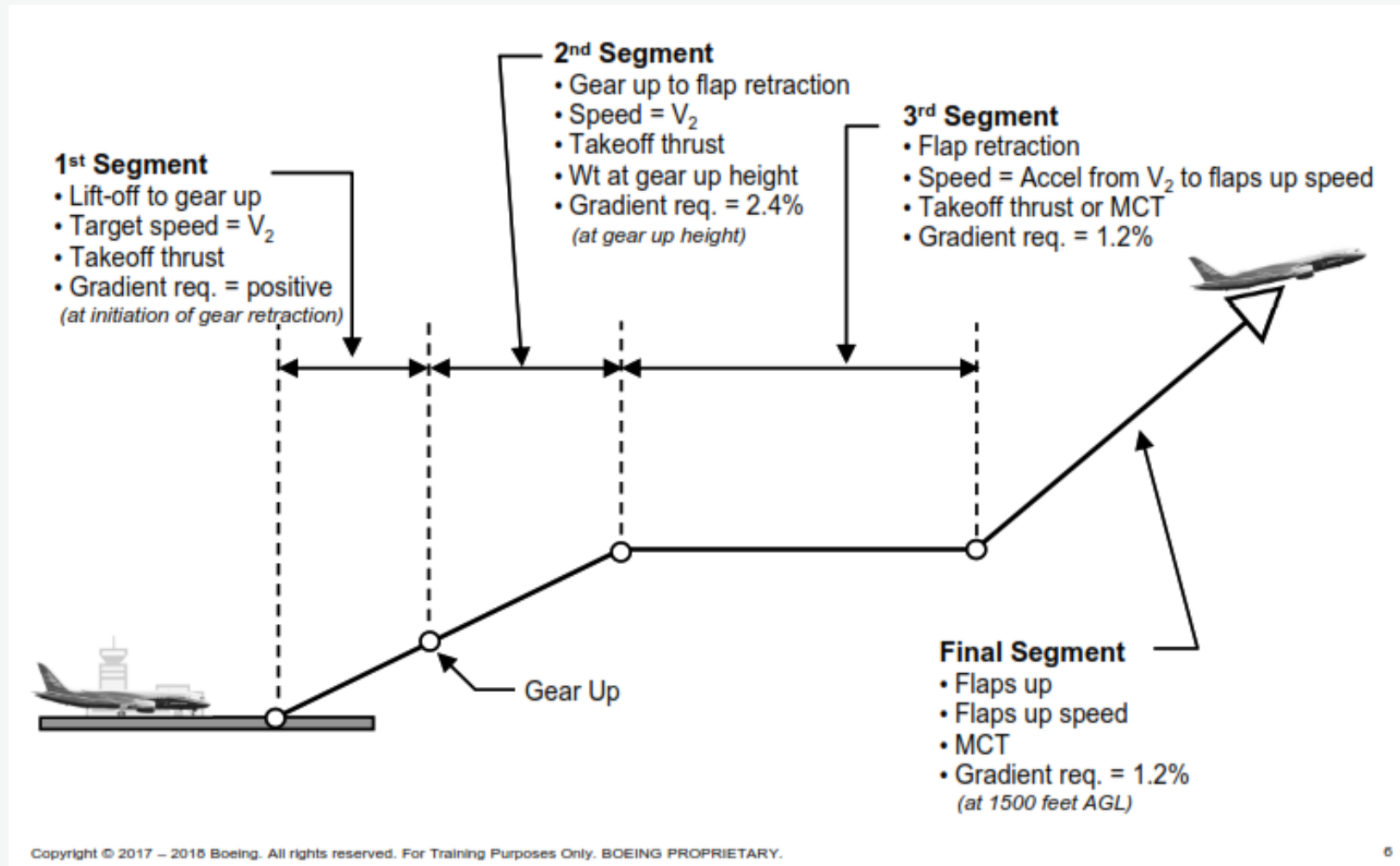


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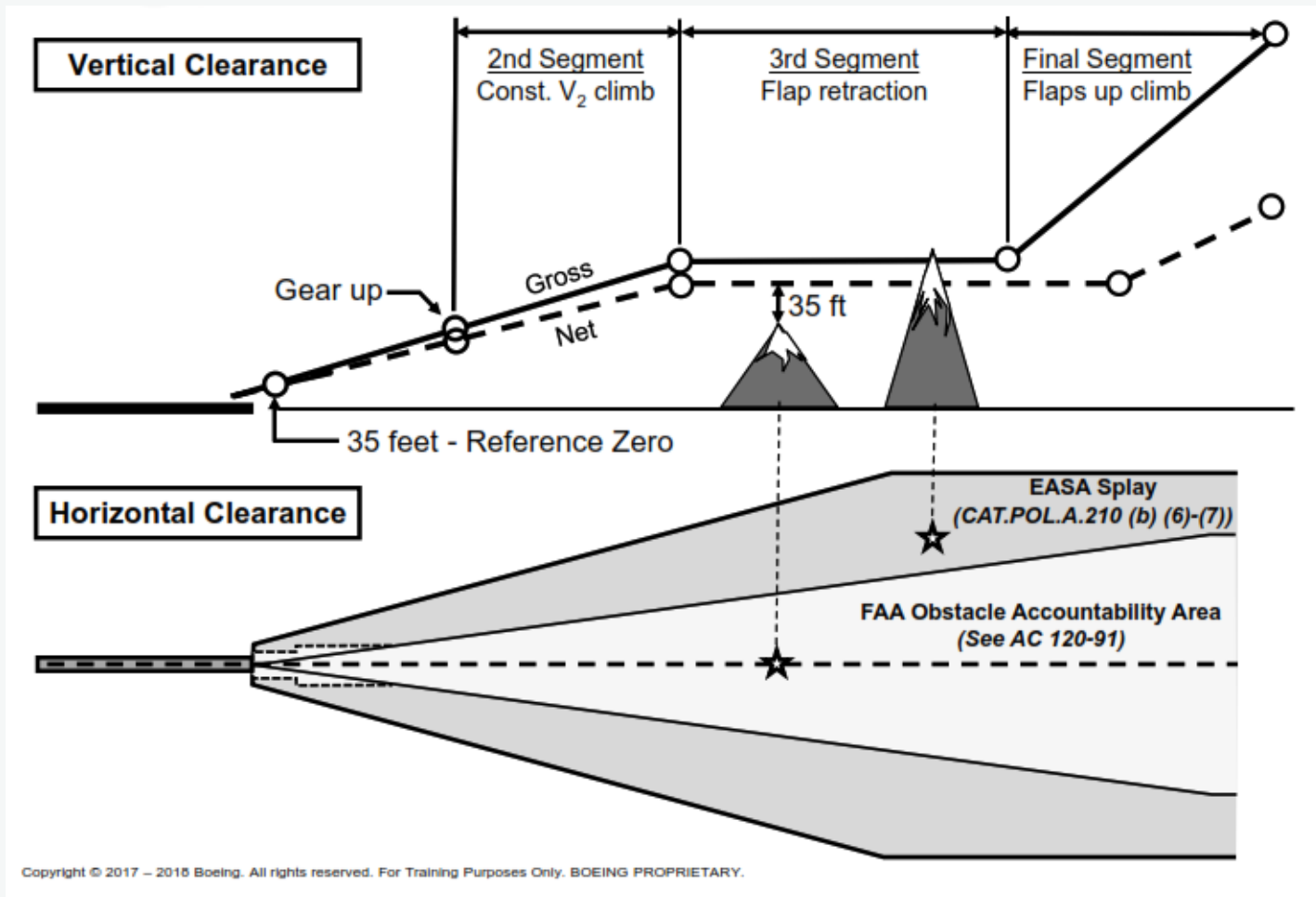
Takeoff Performance

One Engine Inoperative Takeoff Flight Path



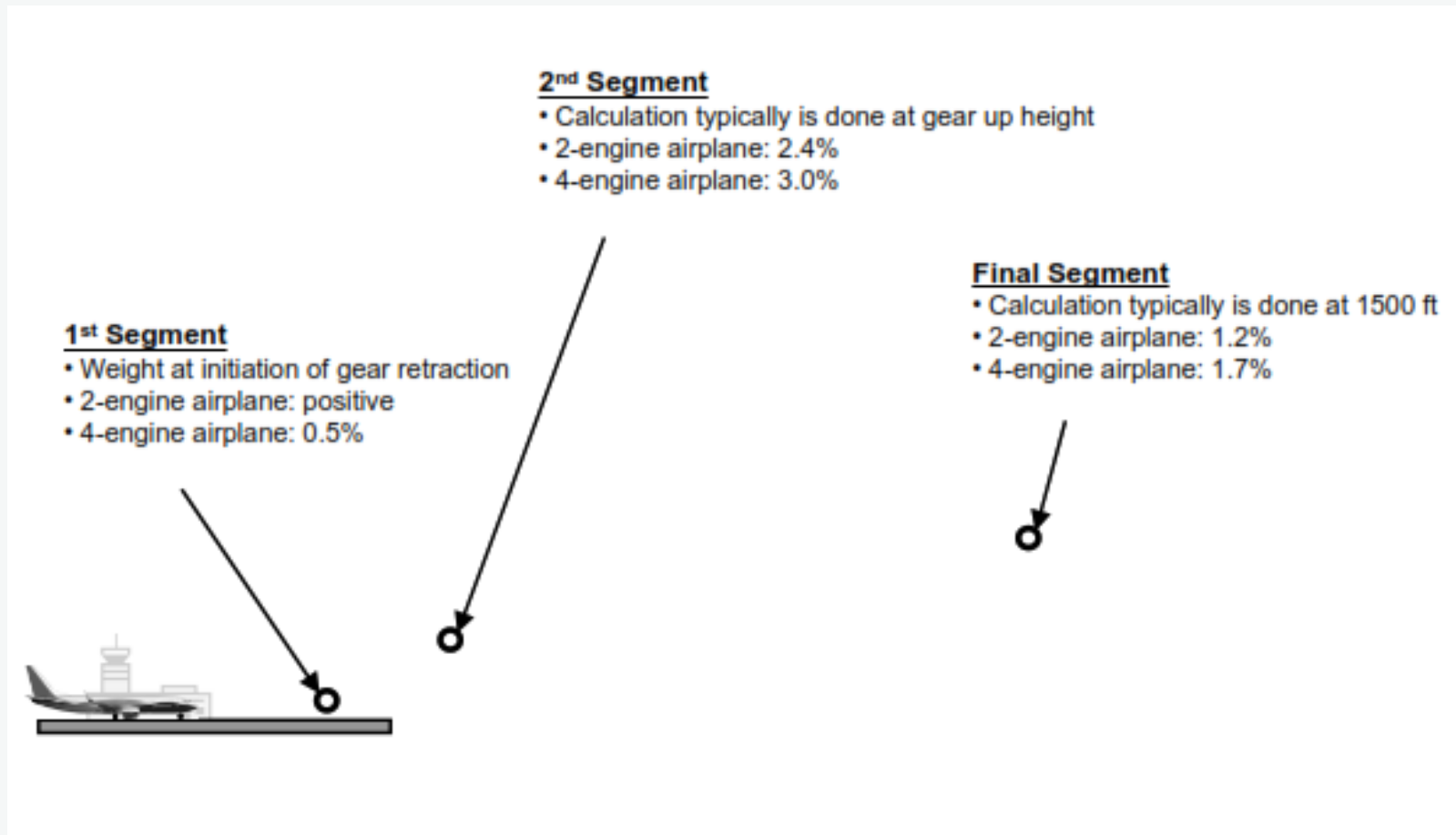
Takeoff Performance

One Engine Inoperative Takeoff Flight Path Obstacle Clearance Requirements



Takeoff Performance

One Engine Inoperative Minimum Climb Gradient Requirements



Takeoff Performance

Tire Speed / Brake Energy Limits

Tire Speed Limit

- 787 models are certified with 235 mph tire speed ratings (one-time over-speed of 260 mph).
- Airplane lift-off speed must be less than the tire limit speed to prevent tire damage during the takeoff.

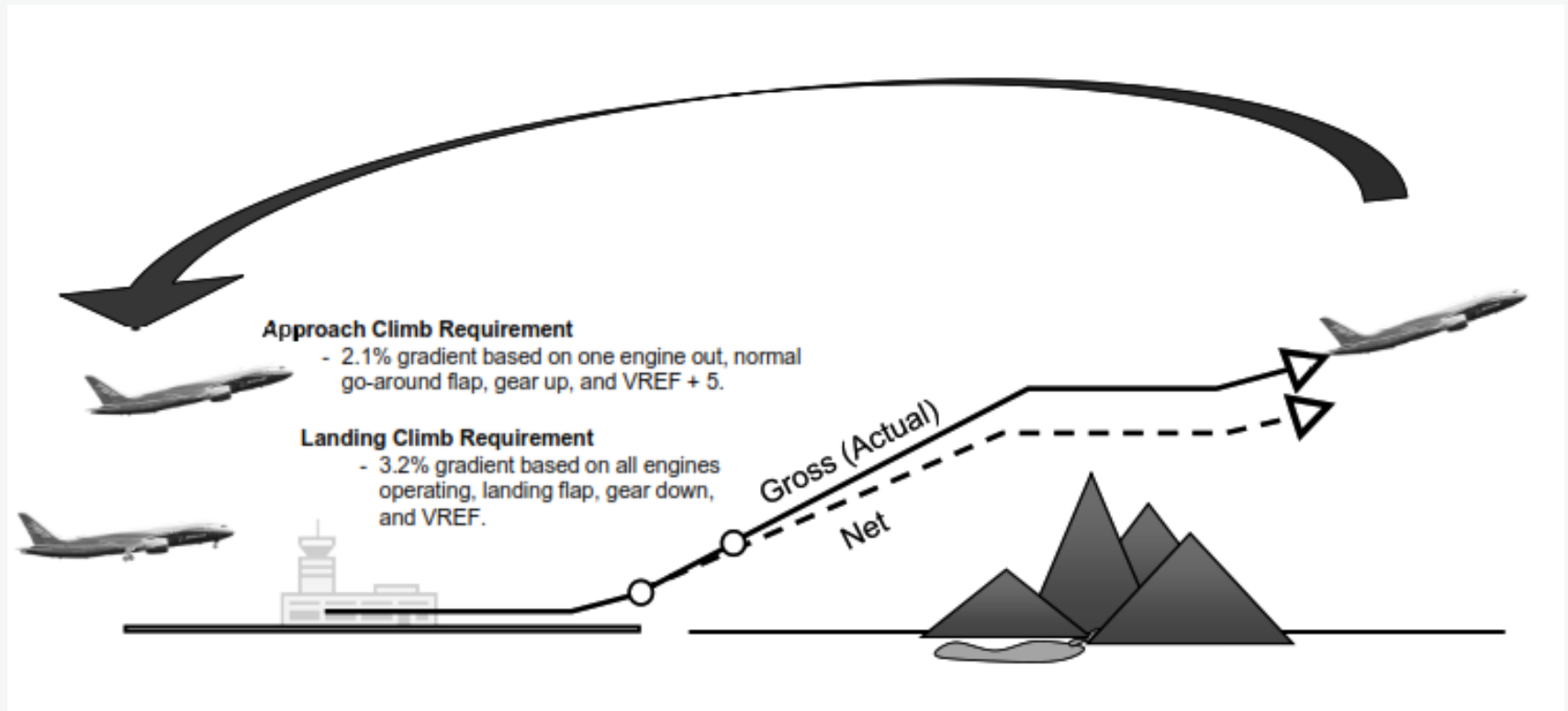
Maximum Brake Energy Speed, VMBE

- VMBE is the highest takeoff rejection speed from which the airplane may be brought to a stop without exceeding the maximum energy absorption capability of the brakes.
- Maximum Brake Energy Speed is compared in takeoff planning to V1 speed. V1 must be less than VMBE for a safe takeoff

Takeoff Performance

Return-to-Land Requirement

The airplane must be able to meet the same minimum go-around gradients that are required for dispatch landing performance



Takeoff Performance

Return-to-Land Assumptions and Impact

The go-around gradients are based on takeoff weight with fuel burn credit for the following:

- Takeoff and climbout to 5000 ft
- Level flight / hold for 30 minutes
- Fuel jettison for 15 minutes
- Descent and approach

The maximum allowable takeoff weight may be limited by the Return-to-Land requirements under these conditions:

- High / hot airports
- Low takeoff thrust rating (even at sea level)
- High zero fuel weight (ZFW)
- Low fuel load

Takeoff Performance

Wet Runway Takeoff Performance

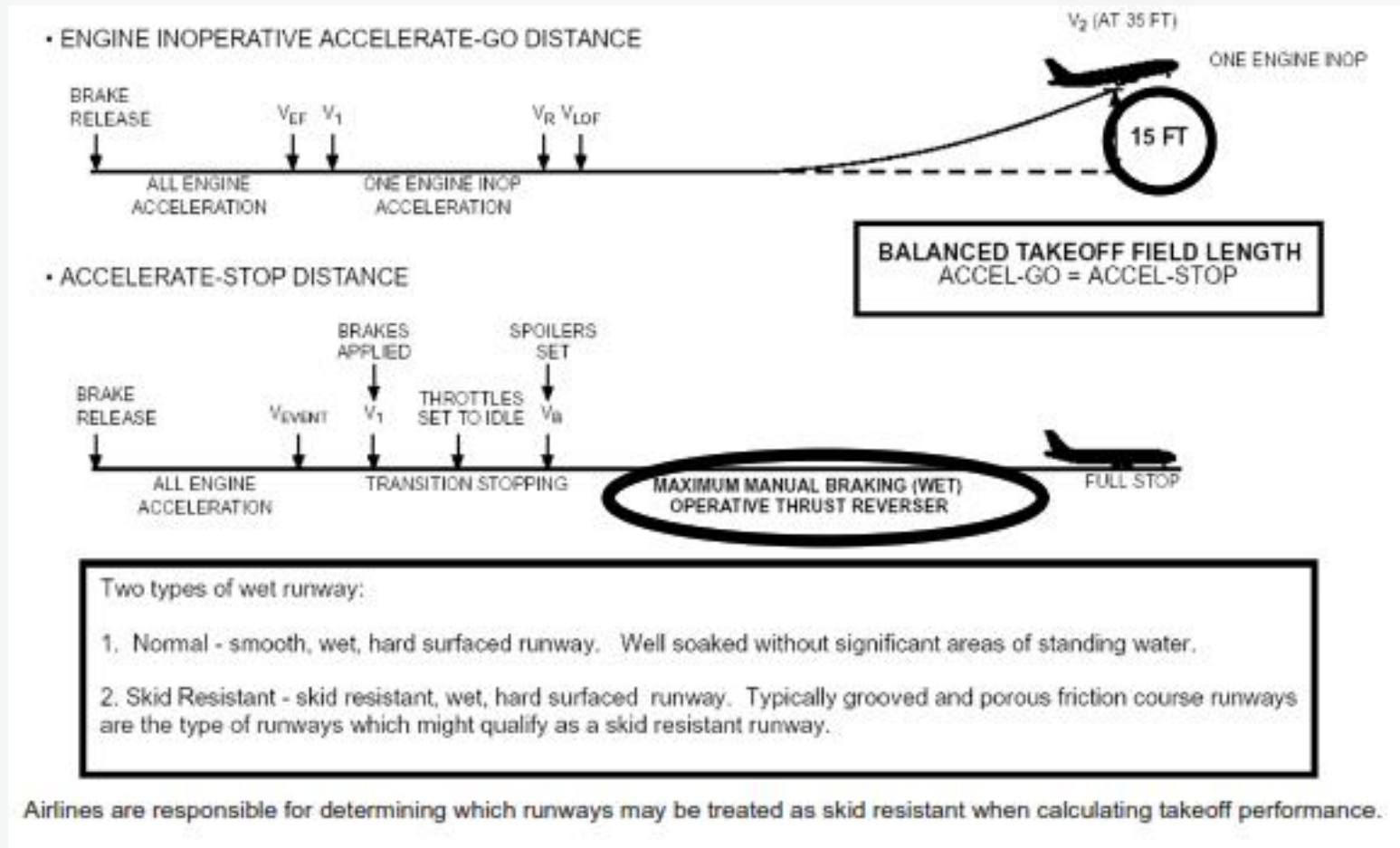
Introduction

- When the runway is wet, takeoff performance must be calculated considering the effects of reduced braking capability.
- Regulations allow credit for use of operative thrust reversers and a screen height of 15 feet when making the takeoff calculation.
- Because the reduced screen height is allowed, credit for clearway is not permitted and obstacle clearance margin is reduced.

<u>Dry Field</u>	<u>Wet Field</u>
No thrust reverser	Available thrust reversers
35 feet screen height	15 feet screen height
Dry runway braking coefficient	Wet runway braking friction
Clearway allowed	Clearway not allowed

Takeoff Performance

Wet Runway Takeoff Performance



Takeoff Performance

Slippery and Contaminated Runway Takeoff Performance

Slippery Runway

Boeing provides data as a function of reported braking actions of: Good, Good-to-Medium, Medium, Mediumto-Poor, and Poor.

FAA and EASA advisory material permit the operator to take credit for use of operative thrust reversers and a screen height of 15 feet when making the takeoff calculation. Because the reduced screen height is allowed, credit for clearway is not allowed and obstacle clearance margin is reduced. The data is presented as a function of "Reported Braking Action". The following is a qualitative description of the terms.

"Good" - Approximately half of ($1/2$) the dry runway capability. Historically half of the dry has been accepted as a wet runway by many certification agencies. Recent JAR certifications has accepted the braking performance equivalent to good for a compacted snow covered runway.

"Poor" - Approximately one eighth ($1/8$) of the dry runway capability. Recently JAR certifications has accepted the braking performance equivalent to poor for wet ice covered runway.

Takeoff Performance

Slippery and Contaminated Runway Takeoff Performance

Slippery Runway

<u>Dry Runway</u>	<u>Slippery Runway</u>
No thrust reverser	Available thrust reversers
35 feet screen height	15 feet screen height
Dry runway braking coefficient	Slippery runway braking friction
Clearway allowed	Clearway not allowed

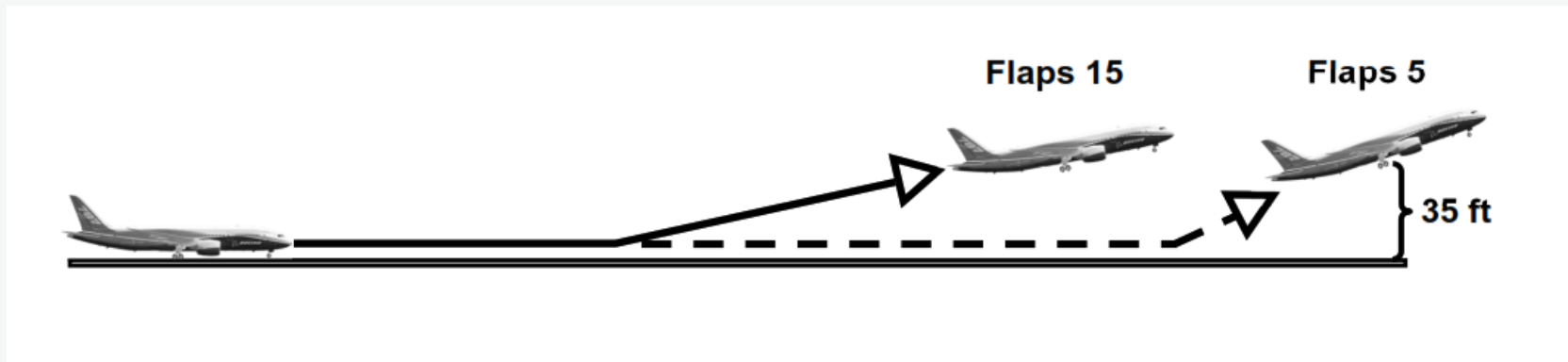
Contaminated Runway

A contaminated runway is a runway covered with a measurable depth of slush, standing water or snow. A contaminated runway degrades the airplane's ability to accelerate and to stop. The performance data for a contaminated runway is calculated based on the same performance assumptions as a slippery runway (4.5m screen height and available reverse thrust).

Takeoff Performance

Effect of Flap Setting on Field Length

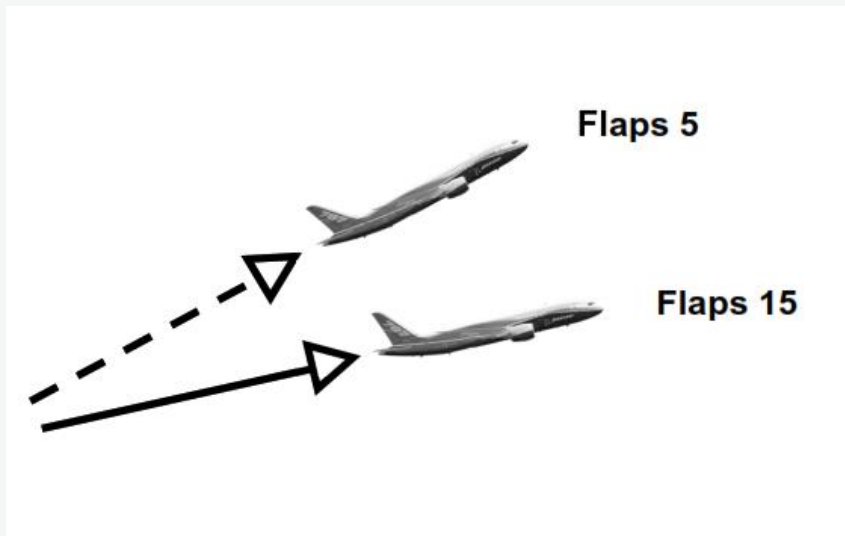
- Flaps setting affects field length required for takeoff and accelerate-stop distances
- A higher number flaps setting (more deflected wing surface) produces the required lift at a lower liftoff speed, resulting in shorter takeoff distance.
- The lower liftoff speed allows a lower V_1 for a balanced field length, resulting in a shorter accelerate-stop distance.



Takeoff Performance

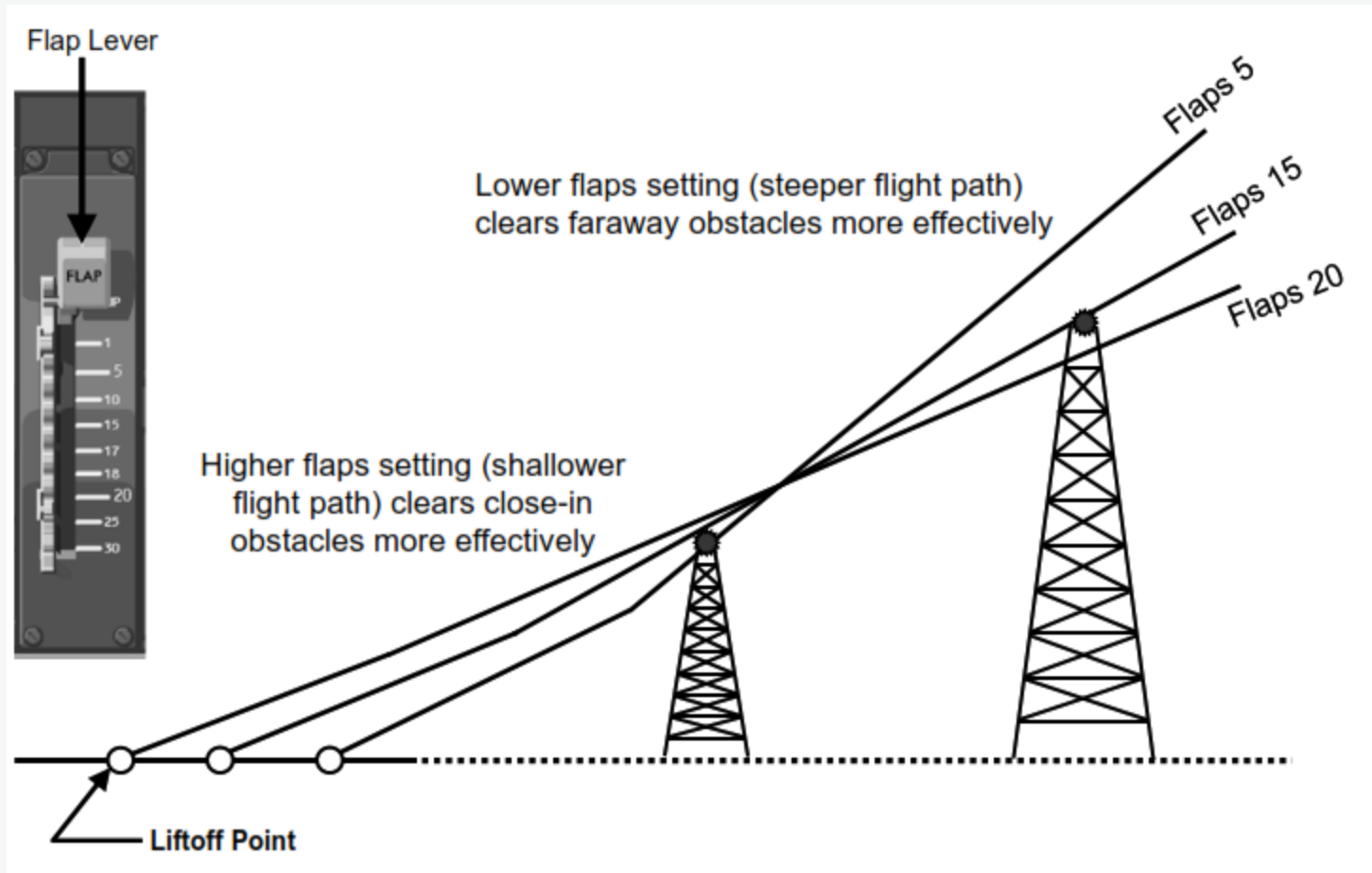
Effect of Flap Setting on Climb Gradient

- Flaps setting affects climb gradient capability.
- Lower number flap settings (less deflected wing surface) use higher flight speeds (V_2), and lower airplane pitch attitudes
- This results in less drag, higher Lift/Drag, and higher climb gradient



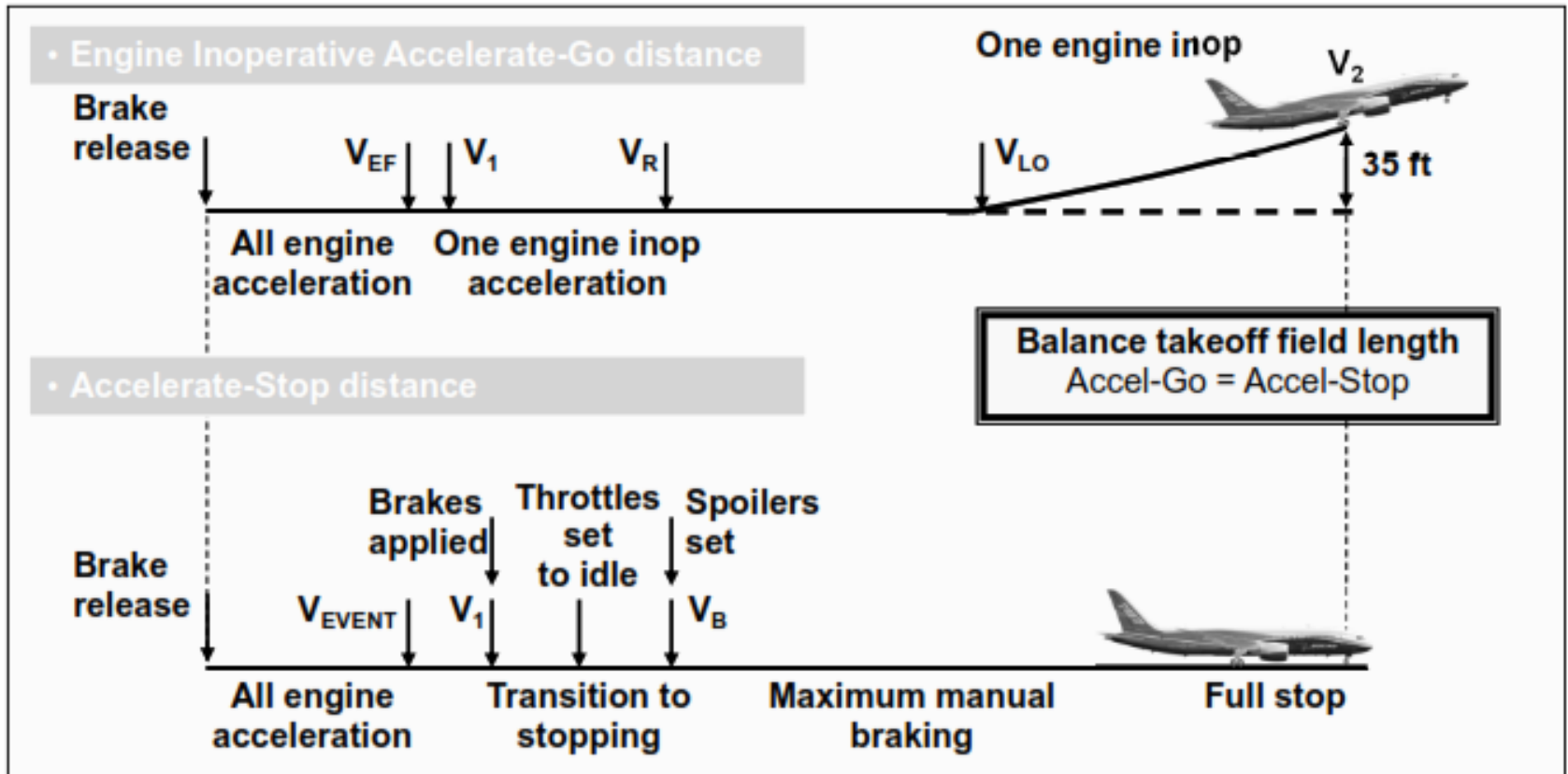
Takeoff Performance

Effect of Flap Setting on Obstacle Clearance



Takeoff Performance

Optimizing V1 Balanced Field Length



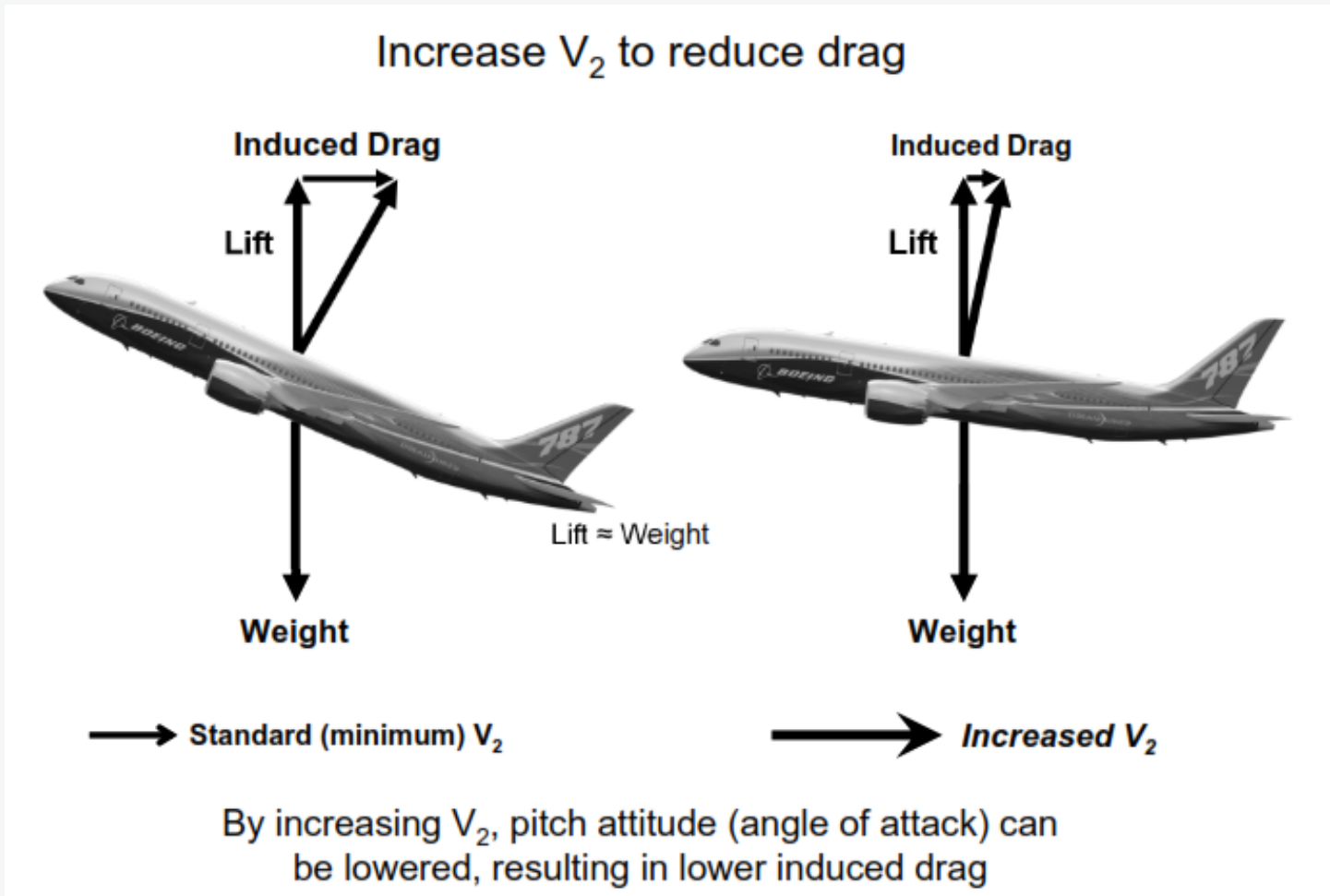
Takeoff Performance

Optimizing V2 (Improved Climb)

- Improved Climb can be used to increase maximum allowable takeoff weight when it is limited by either second segment climb or clearing a distant obstacle.
- Boeing's basic takeoff speed schedules are designed to minimize runway required. When excess field length is available, it can be used to accelerate to a higher VR. The higher VR will result in a higher V2, which allows a lower pitch attitude (angle of attack) and, therefore, results in lower drag. The lower drag allows a higher takeoff weight to achieve the same 2.4% gradient requirement, or a higher flight path angle to better clear distant obstacles.
- When Improved Climb is used, the regulatory requirements for field lengths (one engine inoperative takeoff distance, accelerate-stop distance, all-engine takeoff distance) as well as tire speed and brake energy limits must still be met at the higher takeoff speeds.

Takeoff Performance

Effect of V_2 on Drag



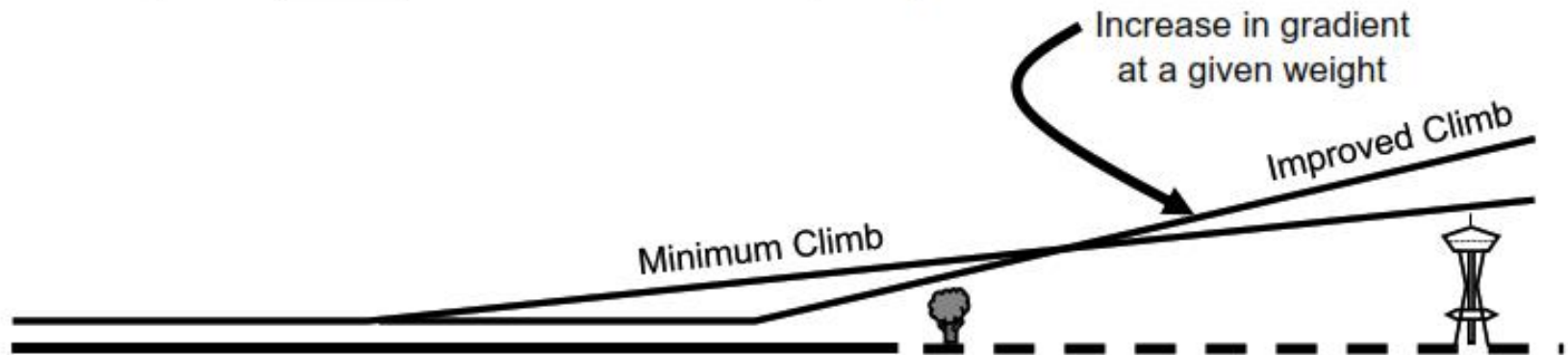
Takeoff Performance

Benefits of Optimized V2 (Improved Climb)

- Increase second segment climb limit weight



- Improves distant obstacle clearance capability



Takeoff Performance

Takeoff Thrust Reduction

Introduction

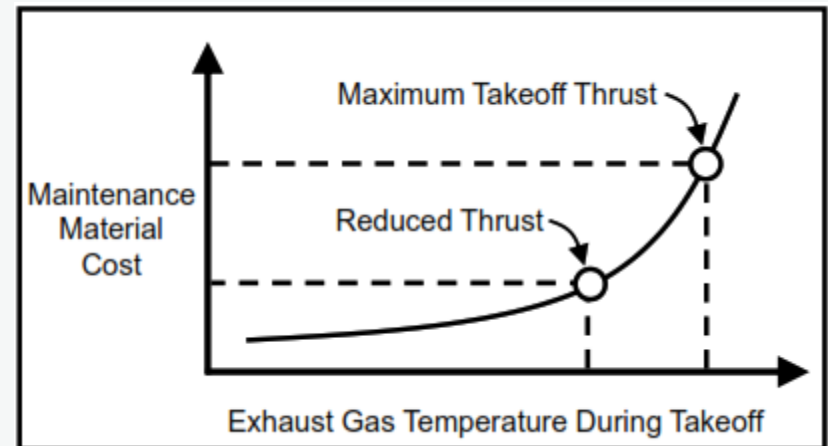
The most wear on the engines occurs during takeoff. It is not necessary to use maximum takeoff thrust when the takeoff is not at maximum allowable takeoff weight. When operating at a thrust level lower than the maximum, the turbine inlet temperature is lowered and the engines' on-wing life increased.

Benefits of using lower takeoff thrust:

- Increased engine reliability
- Increased airplane dispatch reliability
- Decreased engine operating costs
- Decreased engine maintenance costs

Two methods for takeoff thrust reduction:

- Derated Thrust (or Fixed Derate)
- Reduced Thrust (or the Assumed Temperature Method, or ATM)



Takeoff Performance

Assumed Temperature Method (ATM)

The assumed temperature method uses the maximum temperature that meets all the takeoff performance requirements at the expected takeoff weight for the planned takeoff runway.

- The assumed temperature method is conservative. The performance calculation uses a true airspeed based on the assumed temperature. Actual true airspeed for the actual ambient conditions will be lower than the speeds used for the performance calculation. In addition, the actual thrust produced by the engines will be slightly greater than the calculated thrust.

Restrictions

- The maximum thrust reduction permitted is 25% from the chosen rated thrust level.
- The minimum control speeds must be calculated at the normal thrust for the actual outside temperature. This is because the pilot may advance the throttles to the rated takeoff thrust.
- This method is not permitted when the runway is contaminated by water, ice, slush or snow. Also prohibited for dispatch with some systems inoperative.
- This method is permitted on a wet runway if the calculations account for the reduced braking performance.

Takeoff Performance

Derate Selection and Performance Basis

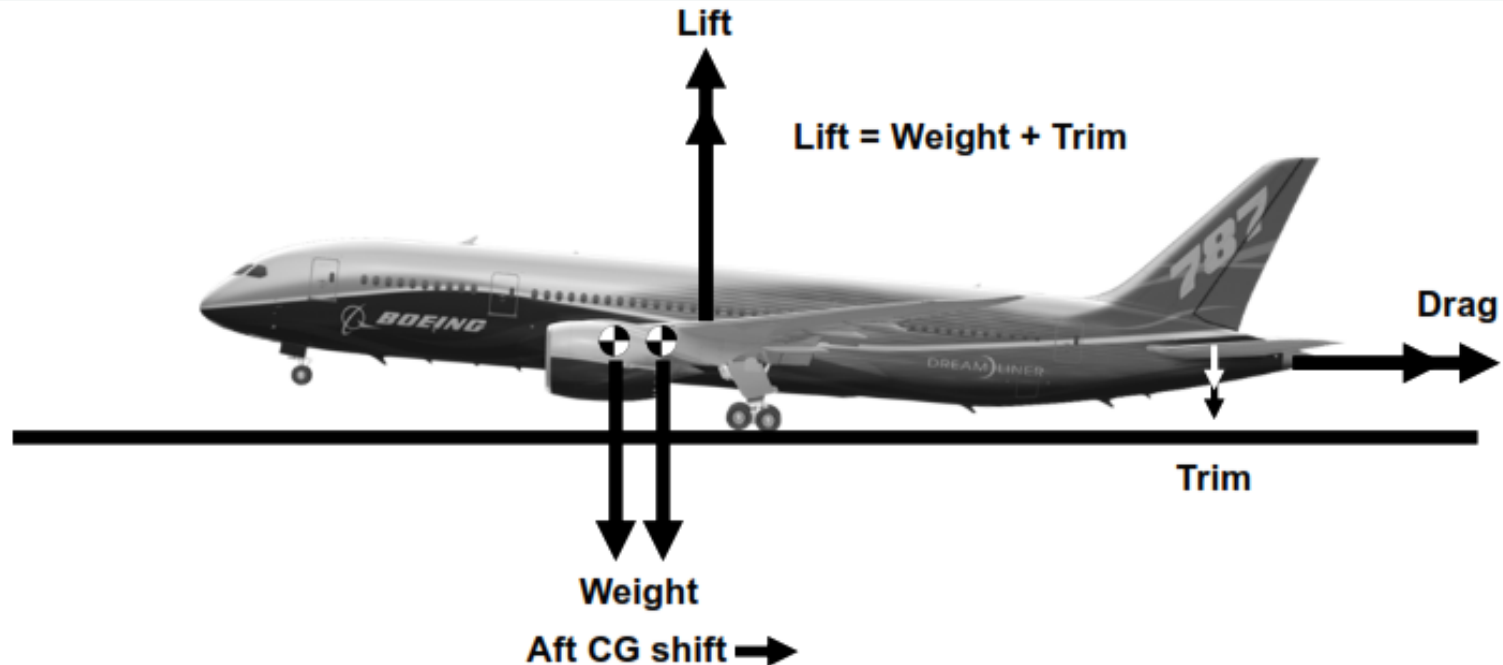
Derated Thrust

- A derate is a certified takeoff thrust rating that is lower than full rated takeoff thrust.
- When operating with a derate, the thrust is considered a limit and should not be exceeded. Therefore, minimum control speeds (VMCG and VMCA) are lower because they are based on the lower thrust. As result, using a derate can increase the maximum allowable takeoff weight when it is limited by those control speeds, such as when takeoff on a runway that is short or contaminated with standing water, slush, or ice.
- Takeoff derates are selected, and changeable, by the operator for the 787.
- Each of the two takeoff derates can be selected from 1% to 40% reduction from the full rated takeoff thrust.
- Boeing defaults at aircraft delivery:

TO 1 = -10% and TO 2 = -20%

Takeoff Performance

Effect of CG on Performance



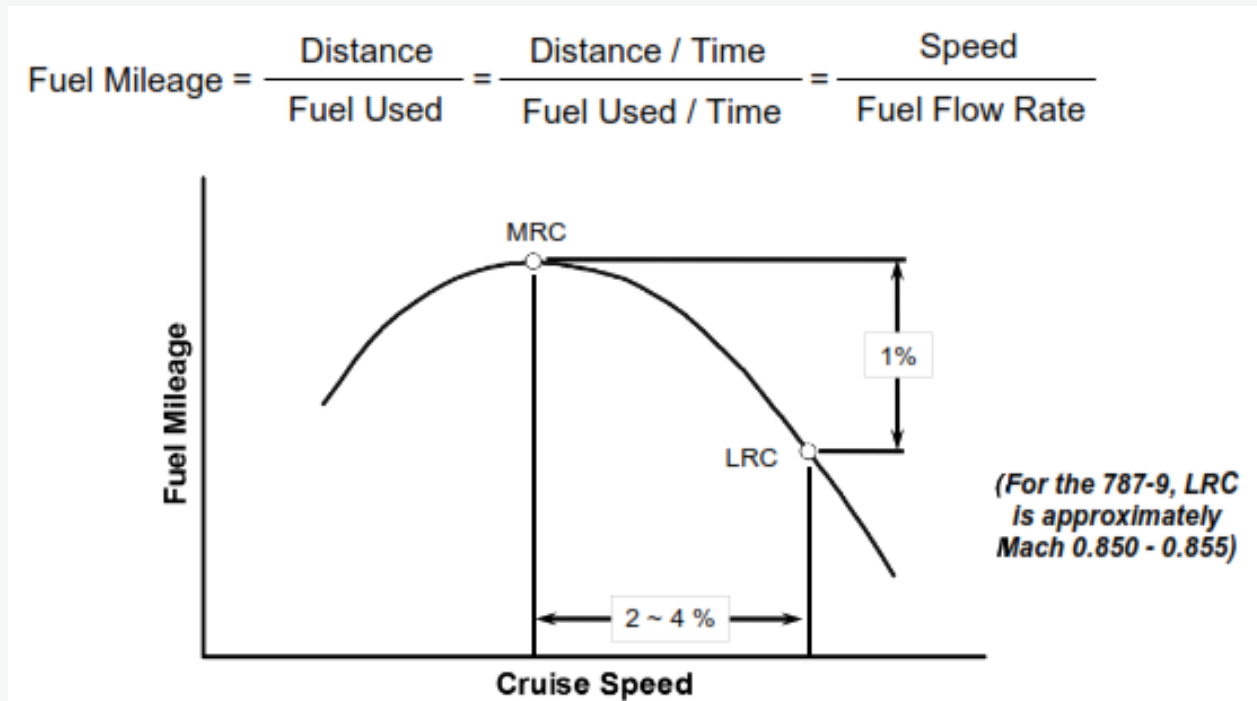
- Reduced drag:
 - ✓ Increased climb limited weight or obstacle clearance limit weight
- Reduced liftoff speed → Lower VR:
 - ✓ Increased field length limit weight or obstacle clearance limit weight
 - ✓ Increased tire speed limit weight
- Reduced stall speed → Lower V2:
 - ✓ May increase field length limit weight or obstacle clearance limit weight
 - ✓ May **reduce** climb limited weight or obstacle clearance limit weight !

Optimum Altitude

Optimum Altitude

Speed and Altitude Selections

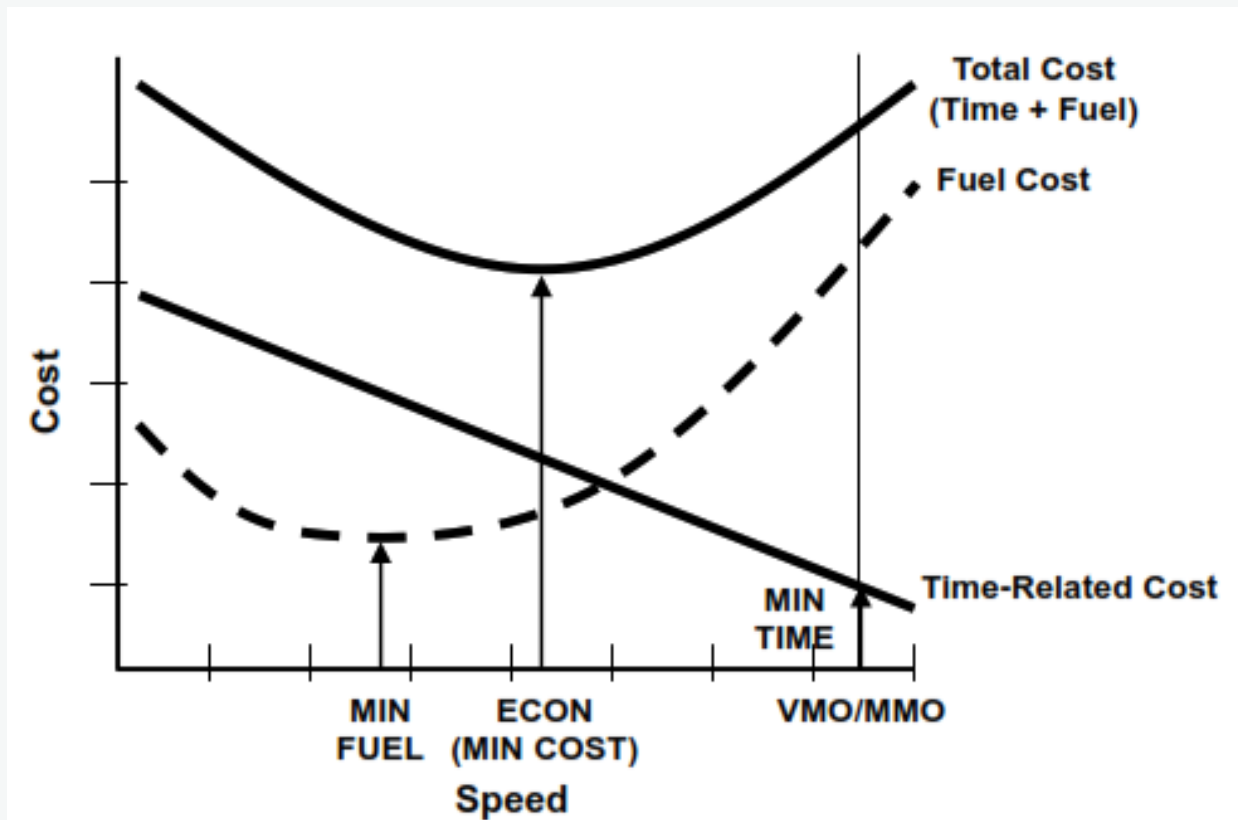
Long Range Cruise (LRC) is the speed above the Maximum Range Cruise (MRC) speed that gives 99% of maximum fuel mileage. The 1% reduction in fuel mileage returns a 2% to 4% increase in cruise speed.



Optimum Altitude

Speed and Altitude Selections

Economy Speed is the speed which minimize the total trip cost (time + fuel)



Optimum Altitude

Speed and Altitude Selections

Cost Index is the relative importance of time cost compared to fuel cost.

$$\text{Cost Index} = \frac{\text{Time-related Cost (dollar / flight hour)}}{\text{Fuel Cost (cents / pound)}} \quad \frac{(\text{currency / flight hour})}{((\text{currency / pound}) \times 100)}$$

- High cost index value means time related cost is more critical than fuel cost. Therefore, flight speed will be faster to reduce trip time.
- Low cost index value means fuel cost is more critical than time related cost. Therefore, flight speed will be slower to save fuel.

Example:

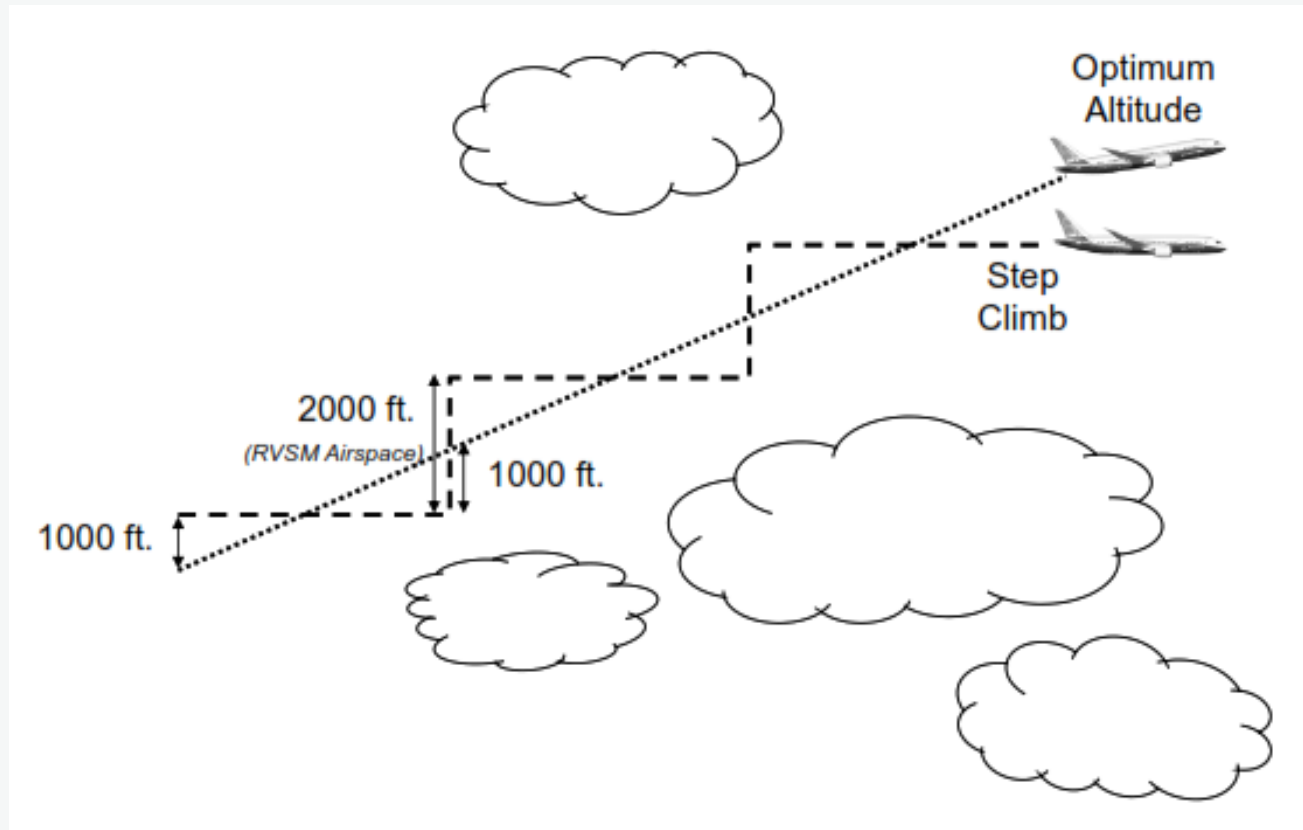
Time related cost is 10000 dollars per hour
Fuel cost is 50 cents per pound

$$\text{Cost Index} = \frac{10000}{50} = \mathbf{200}$$

Optimum Altitude

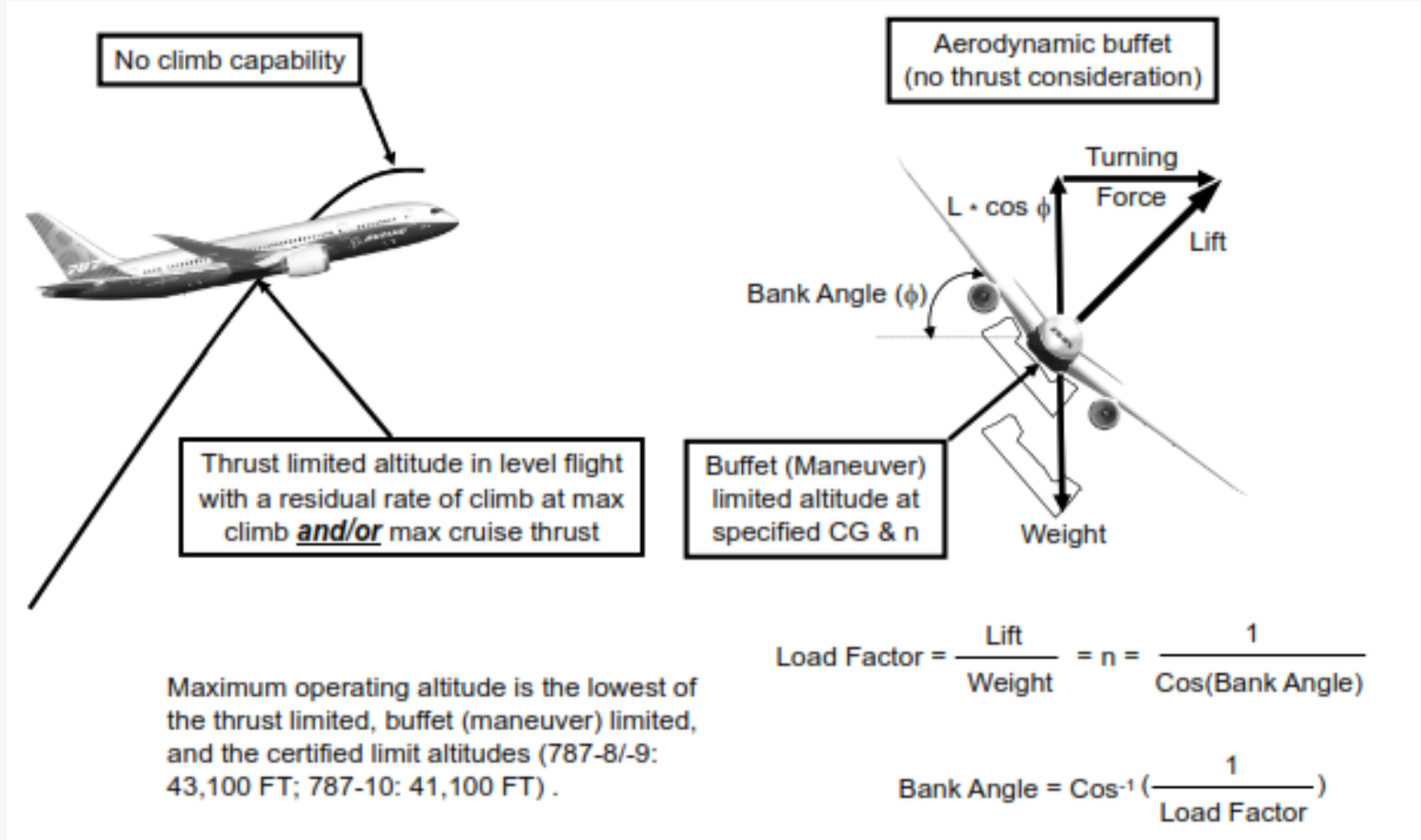
Speed and Altitude Selections

Step Climb to Stay Near Optimum Altitude

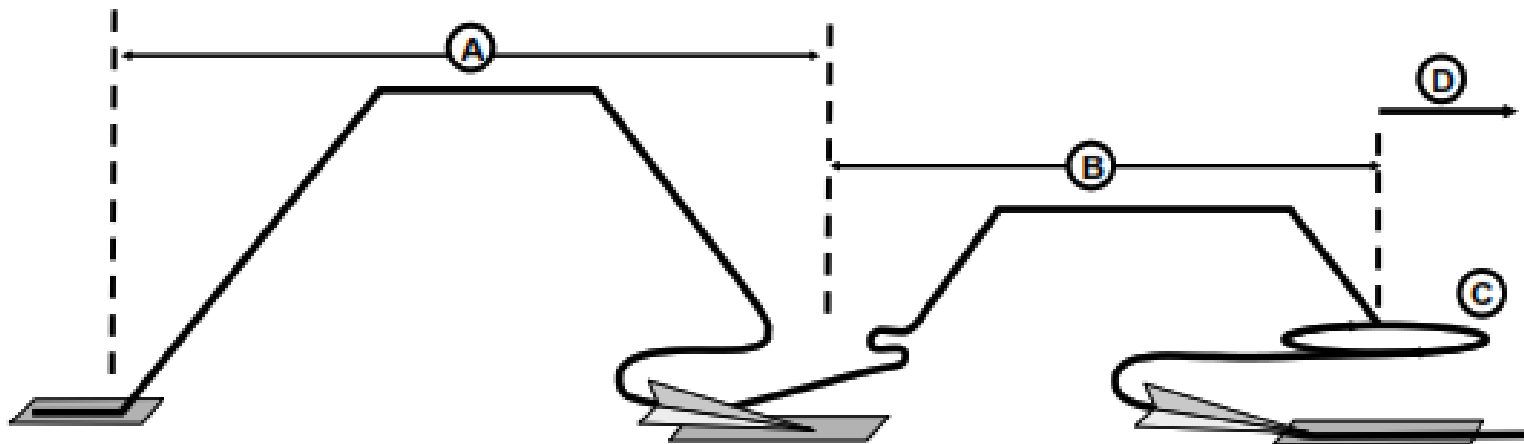


Optimum Altitude

Speed and Altitude Selections - Altitude Capability



Optimum Altitude

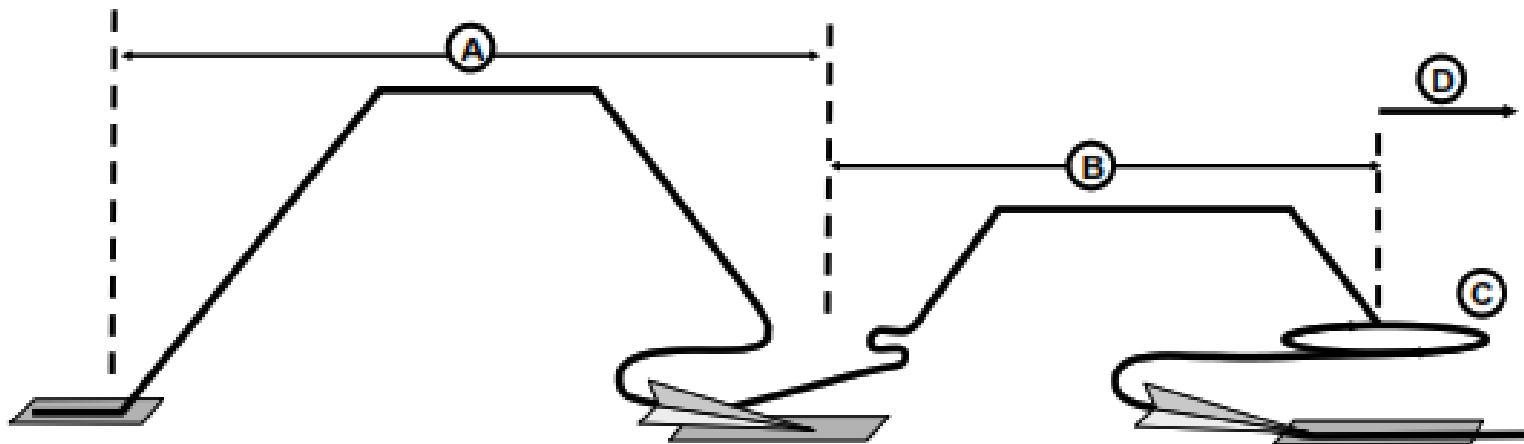


The airplane must have enough fuel:

- (A) To fly to and execute an approach, and a missed approach, at the aerodrome to which the flight is planned, and thereafter;
- (B) To fly to the alternate aerodrome specified in the flight plan; and then
- (C) To fly for 30 minutes at holding speed at 1500 ft above the alternate aerodrome under standard temperature conditions, and approach and landing; and
- (D) To have an additional amount of fuel sufficient to provide for the increased consumption on the occurrence of any of the potential contingencies specified the operator to the satisfaction of the state of the operator (typically 3% to 6% of the trip fuel).

Fuel planning

Optimum Altitude



The airplane must have enough fuel:

- (A) To fly to and execute an approach, and a missed approach, at the aerodrome to which the flight is planned, and thereafter;
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Fuel planning

Optimum Altitude

$$\begin{aligned} &\text{Payload + DOW = ZFW} \\ &\quad + \text{Contingency } (1) \end{aligned}$$

$$\begin{aligned} &= \text{LDG Wt at alternate (wt at end of holding)} \\ &\quad + \text{Holding Fuel } (2) \end{aligned}$$

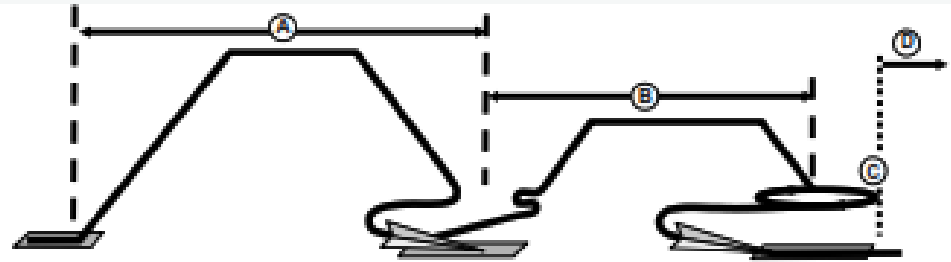
$$\begin{aligned} &= \text{Wt at end of flight to alternate} \\ &\quad + \text{Fuel for flight to alternate } (3) \end{aligned}$$

$$\begin{aligned} &= \text{Wt at end of missed approach} \\ &\quad + \text{Fuel for missed approach } (4) \end{aligned}$$

$$\begin{aligned} &= \text{LDG Wt at destination} \\ &\quad + \text{Trip Fuel} \end{aligned}$$

$$\begin{aligned} &= \text{Takeoff Wt at origin} \\ &\quad + \text{Taxi-out fuel} \end{aligned}$$

$$= \text{Taxi Wt at origin}$$



$$\begin{aligned} &\frac{(1) + (2) + (3) + (4)}{} \\ &= \text{Total Reserve Fuel} \\ &\quad + \text{Trip Fuel} \\ &\quad + \text{Taxi-out fuel} \\ &= \text{Total Fuel Load} \\ &\quad - \text{Taxi-out fuel} \\ &= \text{Takeoff Fuel Load} \end{aligned}$$

**ICAO Fuel Planning
Rules: Fuel & Weight
Accounting Example
Implementation**

Landing Performance

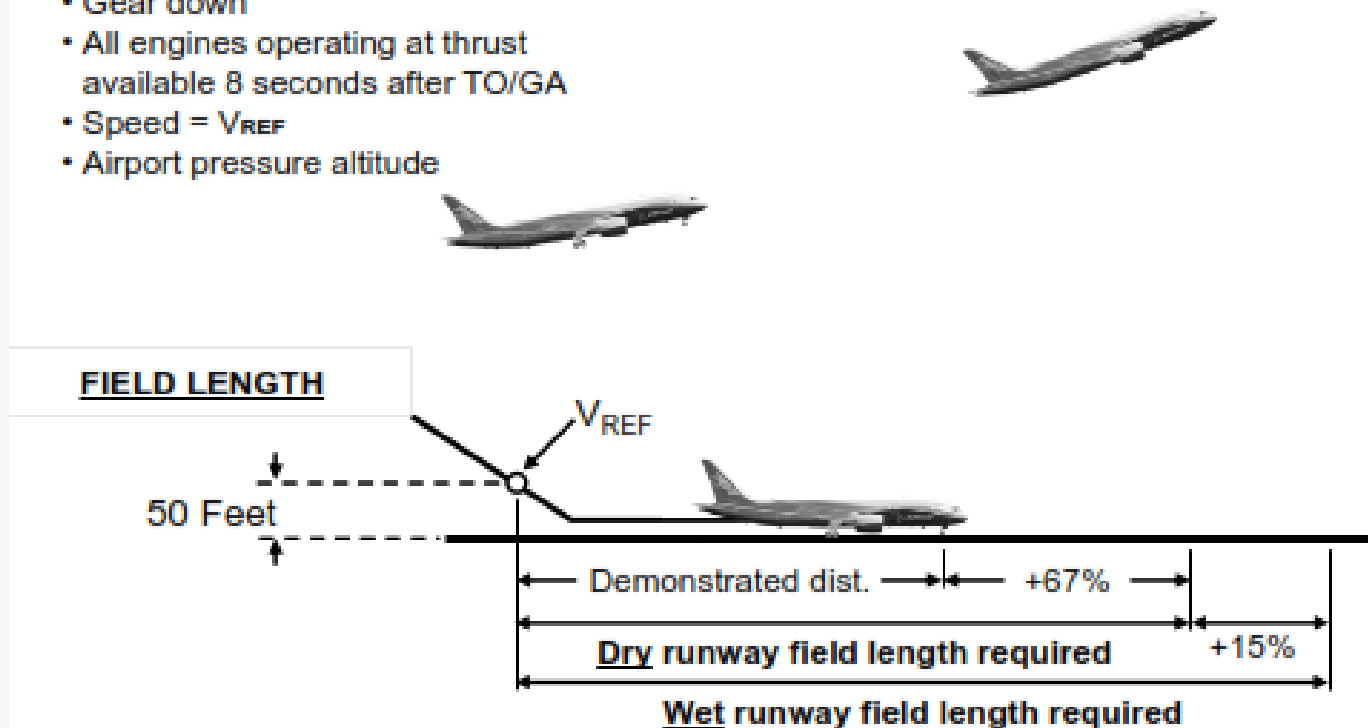
Landing Performance

LANDING CLIMB:

- 3.2% gradient
- Landing flaps setting
- Gear down
- All engines operating at thrust available 8 seconds after TO/GA
- Speed = V_{REF}
- Airport pressure altitude

APPROACH CLIMB:

- 2.1% gradient
- Go-Around flaps setting
- Gear up
- Full go-around thrust with one engine inoperative
- Speed = $V_{REF} + 5$
- Airport pressure altitude



Landing Performance

Dispatch Landing Field Length Required

Demonstrated distance based on:

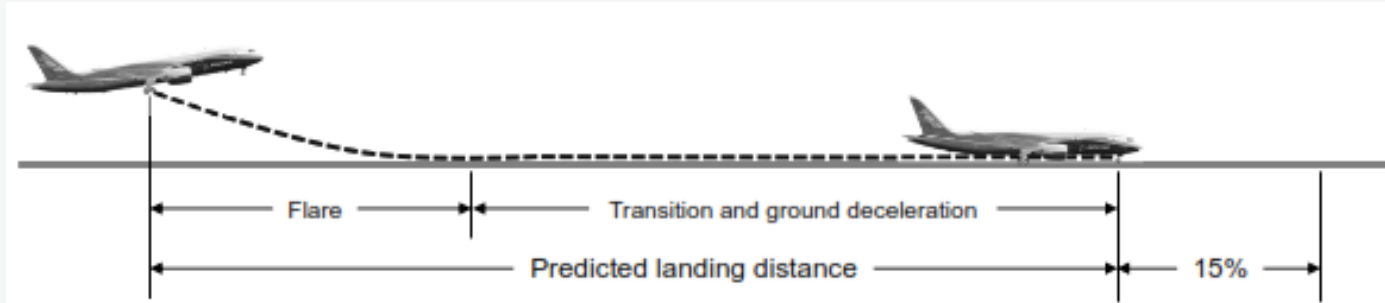
- V_{REF} at 50 feet
- Aggressive touchdown
- Maximum manual braking
- Automatic speedbrakes
- No reverse thrust
- ISA temperature
- No runway slope
- Anticipated pressure altitude and wind

Regulatory factors for dispatch:

- Dry Runway = Demonstrated Distance x 1.67
- Wet Runway = Dry Runway x 1.15
- Slippery Runway (FAA) = Dry Runway x 1.15
- Slippery and Contaminated Runways (EASA) = Actual Distance x 1.15 $\geq$ Wet Runway
- Autoland (FAA) = Manual Landing Distance x 1.15 (= *Wet Runway for 787 models*)
- Autoland (EASA) = Manual Landing Distance + an increment

Landing Performance

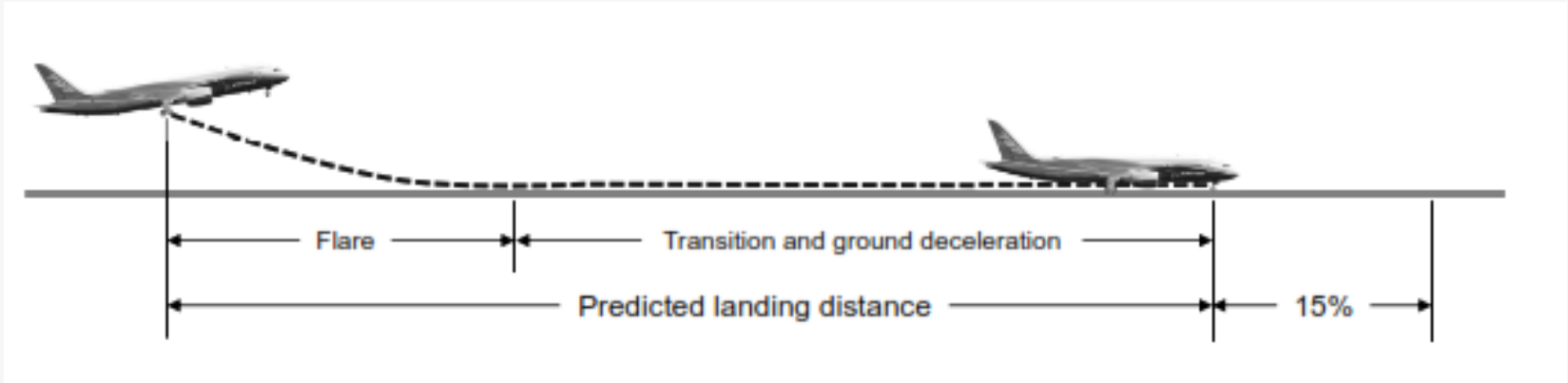
Operational (Time-of-Arrival) Landing Distance



- Landing distance is the horizontal distance from the main gear at the start of flare to the position of the nose gear on the runway where the airplane is stopped.
- Operational distances are typically different from dispatch distances due to differences in actual landing conditions and landing execution relative to those assumed in the dispatch calculations.
- FAA strongly recommends a minimum 15% factor be applied to predicted distances in the determination of required operational landing distances.

Landing Performance

Factors Affecting Actual Landing Distance



- Airplane gross weight and landing flaps setting
- Pilot techniques (flare and touchdown)
- Glide slope angle
- Approach speed and height over runway threshold
- Autobrake setting, speedbrake, and reverse thrust
- Runway surface condition
- Runway slope
- Ambient condition (wind, temperature, and pressure)

Thank you.

